FoodVerse AI - Project Report

1. Abstract

FoodVerse AI is a multimodal food intelligence platform. It analyzes food images, voice and text to provide nutrition, freshness, hygiene, cooking and personalized meal insights.

2. Introduction

Short background on why food analysis matters (health, safety, allergies).

3. Problem statement

See `1. Brainstorming & Ideation/problem_statement.md`.

4. Objectives

- Identify dishes and estimate nutrition from a photo.
- Detect food spoilage.
- Monitor hygiene.
- Help with cooking.
- Generate safe, personalized meal plans.

5. System design

See `3. Project Design Phase/`.

6. Implementation

Frontend: React. Backend: FastAPI. Models: PyTorch and OpenCV. LLM: Meta Llama through Ollama. Details in `5. Project Development Phase/README.md`.

7. Testing and results

See `6.Project Testing/`. Freshness model: 77.8% on own photos (21 of 27); best confidence threshold 0.5 with 85.2% overall.

8. Limitations

- Nutrition values are estimates.
- The recognition model only knows the dish classes it was trained on.
- The meal planner needs a local Ollama model and enough RAM.

9. Future scope

- More dish classes and regional cuisines
- Portion size estimation from images
- Mobile app
- Wearable and health-app integration

10. Conclusion

FoodVerse AI combines food recognition, freshness detection, a hygiene checklist, a smart kitchen and an AI meal planner in one web app. On 27 own test photos the freshness model reached 77.8% accuracy, and 85.2% with the best confidence threshold of 0.5. The project was built solo between 16 and 23 October 2026.

11. References

- PyTorch: https://pytorch.org
- FastAPI: https://fastapi.tiangolo.com
- React: https://react.dev
- Ollama: https://ollama.com
- Meta Llama: https://www.llama.com
- OpenCV: https://opencv.org